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Azure Cost Management and Billing 2021 year in review

By Michael Flanakin, Program Manager, Azure Cost Management

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We started 2021 with great hopes of putting the global pandemic behind us. While it's happening much slower than any of us may have wanted, we're getting there. Yet, while we evolve and learn new ways to work, the need to manage and reduce cost continues to grow. With that in mind, 2021 was more focused on platform capabilities in Azure Cost Management and Billing and the underlying commerce platform at Microsoft. You saw many improvements, but there are even more changes behind the scenes that are extending the foundation for many great things to come.

Streamlined management behind a single pane of glass

While 2019 laid the foundation for Azure Cost Management and Billing, 2020 closed gaps and expanded management capabilities. 2021 took that even further.

Azure Government had an exciting year with the general availability of Pay-As-You-Go (PAYG) subscriptions in Azure Cost Management and support for cost allocation in February, management group cost exports in May, and the general availability of Azure Government Top Secret in August.

Billing account and subscription management got easier with classic **PAYG** and direct **Enterprise Agreement (EA)** management being fully available from the Azure portal. For EA billing admins, this also comes with new capabilities like visibility into your Microsoft Azure Consumption Commitment (MACC), reservations, and invoices. If you manage a direct EA account, you should be working *exclusively* in the Azure portal.

Last year, we shared our plans to deliver more capabilities on **Microsoft Customer Agreement (MCA)**. While we covered invoice management updates in October, we haven't shared much about the capabilities that are actively being rolled out or in early preview stages, like Sponsorship support and the PAYG transition to MCA. We also started piloting the faster usage data pipeline we mentioned last year, which will reduce the time it takes for new usage to be available within Cost Management (currently 8-24 hours). And, lastly, perhaps one of the biggest fundamental changes, Cost Management support for Microsoft 365, Dynamics 365, and more started rolling out in October for CSP partners with customers on MCA.

What's next?

As with 2021, the key theme for 2022 is modernization and transition to MCA. Support for Microsoft 365 and other Microsoft products and services will be expanded to individuals and organizations over time, Sponsorship subscription support in MCA will become generally available, and the full transition of PAYG subscriptions should also be complete. These will bring Cost Management to Sponsorship subscriptions and a more flexible billing model with support for Marketplace purchases and reservations in Cost Management for classic PAYG subscriptions, just to name a few benefits. You'll also see the public rollout of the faster usage pipeline for some accounts. We're very excited to get these into your hands and hear what you'd like to see next.

Rich cost reporting and analytics

Perhaps the most exciting release in the reporting and analytics space from 2021 is the ability to schedule automated emails of your saved cost views from November. This brings cost to your inbox, saving you time every day. Better than that, you can share these emails with others, saving you even more time dealing with custom reporting solutions when a simple email will suffice.

Beyond scheduled emails, you also saw a slew of updates in the cost analysis preview, which was first released in November 2020:

- New views for Resource groups in January, Subscriptions in March, and Services in September.
- Informational cost insights in February and a Labs preview for subscription cost anomalies in October.
- New custom date picker in April with expanded support for relative dates in May.
- Ability to view amortized costs in June.
- New download and export options in August.

Outside of cost analysis, scheduled exports added file partitioning for larger datasets and support for Azure Government management groups in May, and the retail prices API added support for non-USD currencies in April.

What's next?

2022 will continue to expand on analytics and insights capabilities in the cost analysis preview – from a better cost summary with forecast and charts to the full rollout of subscription cost anomalies with opt-in alerting to better representations of your resources, like Azure Kubernetes Service (AKS) and other managed applications. This is only the tip of the spear! Foundational performance, reliability, and usability improvements will make the platform faster than ever to quickly explore your costs within the portal. We're still fleshing out plans, but you can expect to see some highly anticipated updates coming your way in this space.

Flexible cost control that puts the power in your hands

Perhaps one of the most anticipated advancements of 2021 was the addition of budget alerts based on forecasted costs in March. If you haven't updated your budgets to include at least one forecast alert, you should stop reading and go do that right now. Forecast alerts are the best way to stay informed about potential budget overages and give you enough heads-up to address issues to proactively avoid the overage. We're also testing new experiences, like defining a budget when creating new subscriptions and after deploying new resources.

And if you are like many organizations, you probably have shared costs that are great for reducing costs but can sometimes be a pain to charge back to each team using them. Azure Cost Management introduced cost allocation in 2020 to solve that exact problem. Building on top of this, cost allocation was expanded to Azure Government in February and added to all APIs and downloads in May to aid cost accountability outside the portal when you're integrating cost data into your own systems and workflows.

What's next?

Earlier, I mentioned the faster usage data pipeline that's in the early stages today and will be expanded to customers and partners in 2022. As that gets delivered, you'll also see improved budget notification times, with an ultimate goal of alerting you within two hours of either going over or being forecasted to go over your budget. On top of this, you'll also see other advancements in the portal, including the full rollout for budget creation as part of the new subscription form and quick access to create new budgets after deploying resources.

New ways to save and do more with less

While cost savings is often the driving force behind cloud adoption, it is also one of the more elusive goals to achieve. Luckily, Azure delivers new cost optimization opportunities every month on top of the recommendations offered by Azure Advisor. The year was once again loaded with cost optimization opportunities from new and updated offers, reduced pricing, and more.

A few over-arching updates I'd like to call out are our ability to share reservations across subscriptions with management groups and manage Azure Hybrid Benefit for SQL from the portal. If you're using reservations or Hybrid Benefit and you haven't seen these, you're missing out!

On top of these, here's a summary of the services you saw new cost-saving opportunities for in 2021:

- **AI + machine learning**
  - Machine Learning expanded to support Japan West and West US 3.
  - Azure Health Bot expanded to support 8 new regions.
- **Analytics**
  - Data Factory added reservations and expanded to two new regions.
  - Azure Data Explorer released new cost recommendations.
  - HD Insight expanded to Norway East and Brazil Southeast and added support for Dav4-series VMs.
  - Azure Synapse Analytics offered limited-time free quantities.
  - Stream Analytics expanded to 10 new regions in May, September, and October.
  - Azure Purview expanded to Central India.
- **Compute**
  - Virtual machines added a limited preview for on-demand capacity reservations in March, then opened the preview for everyone in September, introduced Azure Hybrid Benefit for Linux, and delivered many size improvements: new Dv5 and Ev5, new Dasv5 and Easv5, DCsv2 reduced price, Ebsv5 and Ebdsv5 preview, HBv2 in UAE North, and HBv3 preview.
  - Cloud services added a new deployment model in preview and general availability.
  - Virtual machine scale sets are now covered by Azure Hybrid Benefit for Linux and supported within Azure Dedicated Host.
  - Azure Dedicated Hosts added support for Intel SGX-based confidential computing VMs.
  - Azure Spot Virtual Machines added new features to improve runtime and simulate evictions.
- **Containers**
  - Azure Kubernetes Service (AKS) added smart defaults and a scale-down mode.
- **Databases**
  - SQL virtual machines can manage Azure Hybrid Benefit and Azure Defender for SQL from the portal, and have free Extended Security Updates.
  - Cosmos DB expanded the free tier, added new cost recommendations, released the Serverless general availability, and previews for cost-optimized diagnostics, integrated cache, and Linux emulator, integrated cache, and Linux emulator.
  - Azure SQL database introduced the ledger preview.
  - Azure Database for MySQL is now available with an Azure free account, added the ability to stop databases to stop incurring charges, supports Flexible Server reservations and auto-grow storage, and expanded Flexible Server to Germany West Central and West US.
  - Azure Database for PostgreSQL is now available with an Azure free account, supports Flexible Server reservations, expanded Flexible Server to eight new regions, and expanded Hyperscale (Citus) to Japan West.
  - Azure Database for MariaDB added the ability to stop databases to stop incurring charges.
- **Hybrid + multicloud**
  - Azure Stack hyperconverged infrastructure (HCI) free trial extended to 60 days.
- **Integration**
  - Logic Apps is now available with an Azure free account.
  - Event Hubs launched the Premium preview.
- **Management**
  - Azure Monitor reduced prices and lowered capacity requirements for dedicated cluster reservations.
  - Application Insights expanded to Japan West and UAE Central.

  - Log analytics expanded to UAE Central, Japan West, and Australia Central 2.
  - Azure Automation expanded to many new regions: Germany West Central, South India, Japan West, Norway East, Switzerland West, UAE Central, and UAE North.
- **Media**
  - Encoding in Azure Media Services is now available with an Azure free account.
- **Networking**
  - ExpressRoute reduced the price for Global Reach and expanded peering locations to Campinas, Dublin 2, Sau Paulo, Denver, Newport (Wales), and Pune.
  - Azure Firewall Premium expanded to Azure Gov Texas, Arizona, and Virginia; and, China East 2 and North 2.
- **Security**
  - Azure Key Vault is now available with an Azure free account.
  - Azure Sentinel reduced prices.
- **Storage**
  - Azure Blob Storage enabled last-access time tracking, offers higher priority rehydration from archive storage, and expanded archive storage to Germany West Central, Norway East, and UAE North.
  - Azure Disk Storage introduced shared disks, added support for changing performance tiers without downtime, reduced prices for Ultra disks, and VM level disk bursting on more VM types in July, then released general availability in November.
  - Azure Backup introduced a limited preview of archive tier for VMs and SQL Server in Azure VMs in March, then general availability in August, and added support for Oracle consistent snapshots.
  - Azure Files introduced premium, hot, and cool storage capacity reservations, increased input or output per second (IOPS) at no charge for Premium tier, and added Server Message Block (SMB) multichannel support.
  - Azure NetApp Files removed the preview waitlist and enabled cross-region replication.
  - Azure App Configuration increased the hourly request limit.
- **Web**
  - App Service reduced prices and released App Service Environment v3 and managed certificates GAs.
  - API Management reduced prices.
  - Azure Maps released a new Gen2 pricing model.
- **Virtual desktop infrastructure**
  - Azure Virtual Desktop added autoscale, a new start VM on connect feature, Windows 11, and Azure China general availability.
  - Azure Stack HCI added support for Azure Virtual Desktop.
  - Azure VMWare Solution introduced the disk pool preview, then added new disk pool capabilities, now has an Azure Migrate assessment, and expanded to Southeast Asia, Australia Southeast, Canada East, Brazil South and East US 2.

What's next?

As usual, you'll see more of the same types of cost optimization opportunities throughout 2021. We'll also continue to partner with service teams to help them deliver cost recommendations and find new ways to help you save more on your existing workloads.

Making it easier to learn and use Cost Management and Billing

We're constantly on the lookout for ways to make Cost Management and Billing easier to learn and use. From ratings and reviews in the portal to user research, like the five surveys we shared in 2021, and many, many conversations with you all – your feedback is critical.

With 14 previews throughout the year, it's hard to pick favorites, but a few of the key updates you should look at are the new custom date picker in the cost analysis preview from February and an update to add relative date ranges in May, then download options and usability updates in the preview in

August, scheduled emails in classic cost analysis and multi-tasking in the cost analysis preview in September, and subscription cost anomaly detection in October.

You also saw usability improvements that make it easier to pick up where you left off and complete tasks quicker, like remembering the last-used scope in January and configuration updates to show billing policy status and common commands for subscriptions and resource groups in August. Of course, these are just a few. We're always looking for ways to improve your experience.

There were also many videos and documentation updates from us, our partners, and the community. It's truly amazing to see how important cost visibility, accountability, and optimization are for everyone from early learners to the largest organizations. We covered 18 videos and 49 of the main documentation updates, but that barely scratches the surface of all the great learning content out there.

What's next?

As always, you can expect to see more of the same in 2022: Continued dedication to ease of use through early access to previews and experimentation. We're especially eager to get your feedback on larger navigation improvements across Cost Management and Billing. We're also currently working on improvements to reporting documentation, both within the portal and via APIs for integration and automation needs.

Looking forward to another year

With everything that happened within the last 12 months, we couldn't list them all here, but I would encourage you to check out and subscribe to the Azure Cost Management and Billing monthly updates for the latest news.

We look forward to hearing your feedback as these new and updated capabilities become available. And if you're interested in the latest features, before they're available to everyone, check out Cost Management Labs and don’t hesitate to reach out with any feedback. Cost Management Labs gives you a direct line to the Azure Cost Management and Billing engineering team and is the best way to influence and make an immediate impact on features being actively developed and tuned for you.

Follow @AzureCostMgmt on Twitter and subscribe to the YouTube channel for updates, tips, and tricks! And, as always, share your ideas and vote up others in the Cost Management feedback forum.

Best wishes from the Azure Cost Management and Billing team.

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